BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Arithmetic, Comparison and Casting

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does assignment do to a variable?

2. What is the CPU's main job?

A) Execute instructions
B) Keep permanent file backups
C) Display every image
D) Supply electrical power

3. A folder contains 30 images of 200 KB each. Using decimal units, give its total size in KB and MB.

4. A photographer travels with large files and needs fast transfers. "The cheapest storage is always best." Correct this. What else matters?

Today's objectives

Apply arithmetic and comparison operators and type conversion. Use arithmetic operators and comparison operators in a practical scenario.

Key vocabulary - Read, cover, recall

Arithmetic operators

Operators such as +, -, * and / allow programs to calculate results.

Comparison operators

Symbols such as ==, < and >= compare values and produce Boolean results.

Casting

int() converts appropriate text or numeric values to integer so arithmetic can use them.

Order of operations

Parentheses can make intended calculation order clear and avoid ambiguous results.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

What is Arithmetic operators?	Arithmetic operators - Operators such as +, -, * and / allow programs to calculate results.
In this example	A train fare calculator uses comparison operators; Symbols such as ==, < and >= compare values and produce Boolean results.
Follow the effect	int() converts appropriate text or numeric values to integer so arithmetic can use them. Explain how this changes the eventual program result.
Check understanding	Order of operations is used for a specific reason: Parentheses can make intended calculation order clear and avoid ambiguous results.
Integer arithmetic and exponentiation	In OCR ERL, 17 DIV 5 is the integer quotient 3, while 17 MOD 5 is the remainder 2. The ^ operator means exponentiation: 2 ^ 3 = 8. Ordinary precedence applies; 4 + 3 * 2 = 10, whereas (4 + 3) * 2 = 14.

I Do - One worked example

A train fare calculator is being written. Explain how arithmetic operators and comparison operators are used to make the program work. [4 marks]

1. We are designing a train fare calculator. Identify which data and decisions are essential.

2. Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1].

3. Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1].

4. Two linked points for each technique give four mark points.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. In a museum ticket price, explain how arithmetic operators helps the program. [2 marks]

2. For a retail discount, explain why comparison operators matters. [2 marks]

3. When designing a sports points, explain how casting and order of operations should be used. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Convert the text "4" into an integer and print how many bits are in four bytes.

```
bytesText = "4"
bytes = ...
print(bytes * 8)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State two programming terms or techniques relevant to this lesson. [2 marks]

Q3. A school trip cost needs to work correctly. Explain how comparison operators and casting would help. [4 marks]

Q4. A pupil is programming a takeaway bill. Explain the roles of arithmetic operators, casting and order of operations in their solution. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Operators such as +, -, * and / allow programs to calculate results. ☐ T ☐ F

2. Arithmetic operators - Operators such as +, -, * and / allow programs to calculate results. ☐ T ☐ F

3. A train fare calculator uses comparison operators; Symbols such as ==, < and >= compare values and produ... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It stores a value in the named variable, replacing its previous value if one existed. ☐

Q2 A - Execute instructions ☐

Q3 6,000 KB = 6 MB. ☐

Q4 Check capacity, speed, cost and how it will be used. An SSD can suit fast transfers and travel, if its capacity and price meet the photographer's needs. ☐

Code First - Model

```
bytesText = "4"
bytes = int(bytesText)
print(bytes * 8)
```

Output: 32

We Do - Check the model

1. Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1]. ☐

2. Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1]. ☐

3. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] - Parentheses can make intended calculation order clear and avoid ambiguous results. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Arithmetic operators [1]; Casting [1]. ☐

Q3 Comparison operators [1] - Symbols such as ==, < and >= compare values and produce Boolean results. [1]. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. ☐

Q4 Arithmetic operators [1] - Operators such as +, -, * and / allow programs to calculate results. [1]. Casting [1] - int() converts appropriate text or numeric values to integer so arithmetic can use them. [1]. Order of operations [1] - Parentheses can make intended calculation order clear and avoid ambiguous results. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: